Maharashtra Education Society's

Institute of Management and Career Courses (IMCC), Pune

AUTONOMOUS

[CGPA-3.32] in Cycle 1

Python

Python Assignment – File Operations

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1. **Write a program to read a file and count how many vowels are present in its content.**

Python Sem-1 > Assignment-1 > question1.py > ...

```
1   # Program to count vowels in a file
2
3   file = open("sample.txt", "r")
4   content = file.read()
5   file.close()
6   count = 0
7   for ch in content:
8       if ch.lower() in "aeiou":
9           count = count + 1
10  print("Total number of vowels:", count)
11
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Admin\OneDrive\Desktop\MCA SEM-1\Python Sem-1> cd Assignment-1
PS C:\Users\Admin\OneDrive\Desktop\MCA SEM-1\Python Sem-1\Assignment-1> py .\question1.py
Total number of vowels: 36
PS C:\Users\Admin\OneDrive\Desktop\MCA SEM-1\Python Sem-1\Assignment-1>
```

Python Sem-1 > Assignment-1 > sample.txt

```
1   This is sample.txt file.
2   This is used for as reference file for programs.
3   Python is easy to learn.
4   File handling is useful.
```

2. **Read all lines from a file into a list. Print the total number of lines and display the 2nd and 4th line.**

Python Sem-1 > Assignment-1 > question2.py > ...

```
1  # Program to read lines from a file
2
3  file = open("sample.txt", "r")
4  lines = file.readlines()
5  file.close()
6  print("Total number of lines:", len(lines))
7  print("Second line:", lines[1].strip())
8  print("Fourth line:", lines[3].strip())
9
```

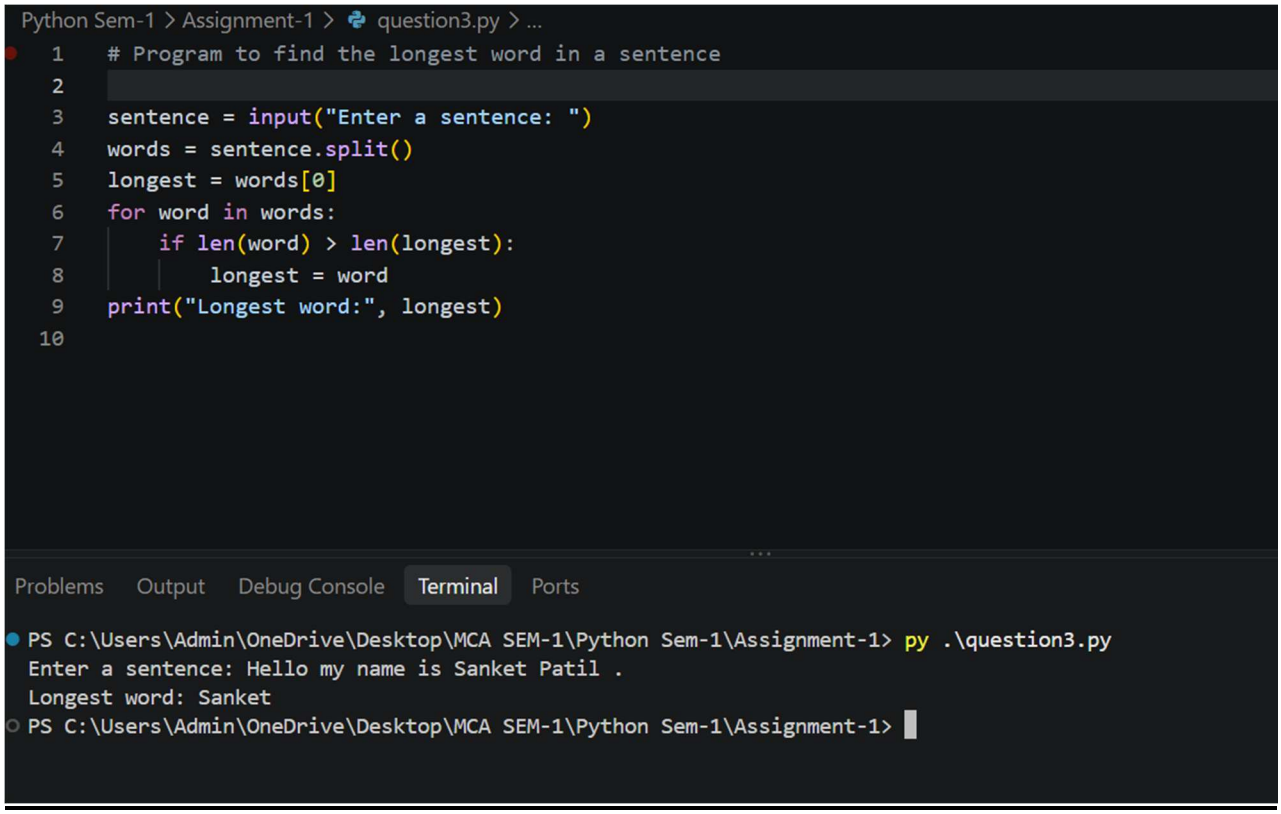
Problems Output Debug Console Terminal Ports

```
PS C:\Users\Admin\OneDrive\Desktop\MCA SEM-1\Python Sem-1\Assignment-1> py .\question2.py
Total number of lines: 4
Second line: This is used for as reference file for programs.
Fourth line: File handling is useful.
PS C:\Users\Admin\OneDrive\Desktop\MCA SEM-1\Python Sem-1\Assignment-1>
```

Python Sem-1 > Assignment-1 > sample.txt

```
1  This is sample.txt file.
2  This is used for as reference file for programs.
3  Python is easy to learn.
4  File handling is useful.
```

3. **Take a sentence from the user, split it into words, and store them in a list. Print the longest word.**



4. Read a file, split its content into words, and store them in a list.

i. Print the total number of words.

ii. Print all unique words.

Python Sem-1 > Assignment-1 > question4.py > ...

```
1   # Program to count total and unique words in a file
2
3   file = open("sample.txt", "r")
4   content = file.read()
5   file.close()
6   words = content.lower().split()
7   print("Total number of words:", len(words))
8   unique_words = set(words)
9   print("Unique words:", sorted(unique_words))
10
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Admin\OneDrive\Desktop\MCA SEM-1\Python Sem-1\Assignment-1> py .\question4.py
Total number of words: 22
Unique words: ['as', 'easy', 'file', 'file.', 'for', 'handling', 'is', 'learn.', 'programs.', 'python', '
reference', 'sample.txt', 'this', 'to', 'used', 'useful.']
PS C:\Users\Admin\OneDrive\Desktop\MCA SEM-1\Python Sem-1\Assignment-1>
```

Python Sem-1 > Assignment-1 > sample.txt

```
1   This is sample.txt file.
2   This is used for as reference file for programs.
3   Python is easy to learn.
4   File handling is useful.
```